

## Data Dictionary

| Term                 | Definition                                                                                             |
|----------------------|--------------------------------------------------------------------------------------------------------|
| Driver               | A road user traveling by private car and interested in parking availability and congestion.            |
| Commuter             | A public-transport user interested in bus arrivals, service diversions, and taxi availability.         |
| Start Location       | The origin point a user begins a trip from (current location or a typed address).                      |
| Destination          | The end point a user intends to travel to.                                                             |
| Map                  | The interactive city map showing transport layers (carparks, bus stops, taxi density, traffic alerts). |
| Carpark              | A parking facility near a destination; may be multi-storey, surface, sheltered, or basement.           |
| Carpark Availability | The number of currently available parking lots at a carpark.                                           |
| Lot Type             | The category of parking lots available (multi-storey/surface/sheltered/basement).                      |
| Bus Stop             | A designated stop with a name and official code where buses pick up and drop off passengers.           |
| Bus Arrival Time     | The next expected arrivals for a bus service at a bus stop.                                            |
| Bus Diversion        | A temporary change to a bus route due to roadworks, incidents, or events.                              |
| Taxi Availability    | The current density or presence of available taxis near a location.                                    |
| Taxi Hotspot         | An area with relatively high taxi availability at a point in time.                                     |
| Traffic Alert        | A notice of a road condition impacting travel ( lane closure, roadwork, accident).                     |
| Lane Closure         | A temporary closure of one or more lanes on a road segment.                                            |
| Roadwork             | Planned maintenance or construction activity affecting road use.                                       |
| Congestion Level     | The observed or reported traffic intensity on a road segment that affects travel time.                 |
| Route                | A path between a start location and a destination, possibly including alternatives.                    |
| Accessibility        | How easily a destination can be reached by bus (stop                                                   |

|                       |                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------|
|                       | proximity, route coverage, transfers).                                                                         |
| Search Query          | User-entered keywords ( location, postal code, bus stop name/code, road name) to find transport info.          |
| Refresh               | A user action that fetches the latest available transport data for items shown on the map.                     |
| Real-Time Data        | Up-to-the-minute information (carpark lots, bus arrivals, taxi availability, traffic alerts) shown in the app. |
| Travel Mode           | A category of transport for a trip recommendation: Car, Taxi, or Bus.                                          |
| Travel Recommendation | The suggested travel mode for a given trip, with supporting reasons.                                           |
| Estimated Travel Time | The expected time to complete a trip under current conditions for each travel mode.                            |
| Estimated Cost        | The expected fare or cost for a mode for the trip.                                                             |
| Profile               | A user's personal information page (name, email) shown after sign-in.                                          |
| Location Permission   | The user's consent for the app/website to access their live location for nearby transport details.             |
| Home Page             | The landing screen after login, providing entry points to map and features.                                    |